

PENN YAN, NY, USA

WMO: 725194

Lat: 42.643N

Lon: 77.056W

Elev: 275

StdP: 98.06

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 54778

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.6	-14.0	-21.7	0.6	-15.6	-19.0	0.7	-12.7	11.7	-1.7	10.6	-1.9	3.5	210	0.508

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	31.6	22.3	29.9	21.2	28.3	20.6	23.6	28.9	22.7	27.8	21.9	26.5	3.9	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.1	17.3	26.4	21.2	16.4	25.3	20.3	15.5	24.6	71.5	28.7	68.2	28.0	65.3	26.5	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
9.3	8.3	7.4	DB	-19.1	34.1	2.4	1.5	-20.9	35.2	-22.3	36.1	-23.7	36.9	-25.5	38.0	
			WB	-19.6	25.0	2.4	0.9	-21.3	25.7	-22.7	26.2	-24.0	26.7	-25.7	27.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.6	-3.7	-2.9	1.7	8.1	14.8	19.5	22.0	21.1	17.5	11.2	5.1	-0.3
	DBStd	10.29	6.31	5.72	6.05	5.22	4.77	3.73	3.07	3.00	4.12	4.85	4.96	5.06
	HDD10.0	1689	425	363	266	97	11	0	0	0	1	45	160	322
	HDD18.3	3546	683	596	516	310	132	31	5	9	63	227	397	578
	CDD10.0	1529	1	1	8	39	158	284	371	343	227	82	13	1
	CDD18.3	344	0	0	0	2	22	65	117	94	39	5	0	0
	CDH23.3	2614	0	0	1	41	212	512	913	633	273	29	0	0
	CDH26.7	754	0	0	0	9	57	152	285	171	76	3	0	0
Wind	WSAvg	3.2	3.9	3.8	3.7	3.6	3.0	2.7	2.6	2.4	2.5	3.1	3.7	3.8
Precipitation	PrecAvg	880	46	45	58	72	78	85	87	86	88	76	66	62
	PrecMax	1069	100	90	156	151	156	146	232	152	145	173	107	108
	PrecMin	716	18	21	14	26	20	24	23	35	38	23	20	26
	PrecStd	92	19	20	30	32	37	36	44	32	32	38	24	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.4	16.0	21.5	27.8	30.8	32.6	33.0	32.8	31.8	26.6	19.6	16.4
		MCWB	11.1	10.5	14.1	17.2	20.1	22.6	23.1	22.8	21.6	19.0	13.3	12.7
	2%	DB	11.0	11.0	17.1	23.5	28.3	30.5	31.5	30.5	28.8	23.5	17.4	11.6
		MCWB	8.2	7.0	11.0	14.6	18.8	21.3	22.6	22.2	21.0	17.1	12.6	8.5
	5%	DB	7.5	7.7	13.8	19.9	25.9	28.2	29.7	28.4	26.5	21.1	15.2	9.1
		MCWB	5.2	4.4	9.6	12.4	17.9	20.5	21.6	21.1	19.8	16.5	10.8	6.7
	10%	DB	4.8	4.9	11.0	17.3	23.0	26.3	28.0	27.0	24.1	18.5	12.8	6.9
		MCWB	2.5	2.2	7.0	11.5	16.2	19.3	20.8	20.4	18.8	14.5	9.3	4.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.9	11.9	14.9	18.6	21.9	24.0	25.1	24.6	23.5	20.8	15.1	12.7
		MCDB	14.0	15.1	19.8	25.9	28.3	29.8	31.1	29.4	28.9	24.1	17.8	15.7
	2%	WB	8.0	7.4	12.5	15.6	20.2	22.6	23.7	23.3	22.2	18.7	13.4	9.4
		MCDB	10.1	10.7	16.5	21.5	26.1	28.4	29.4	28.0	27.0	22.3	16.6	11.3
	5%	WB	5.3	4.9	9.9	13.8	18.9	21.6	22.7	22.4	20.9	16.8	11.7	6.9
		MCDB	7.3	7.6	13.3	19.1	23.9	26.6	27.8	26.8	24.9	20.1	14.6	9.0
	10%	WB	2.8	2.2	7.3	11.8	17.6	20.5	21.8	21.5	19.8	15.1	9.8	4.7
		MCDB	4.7	4.7	10.6	16.4	21.9	24.9	26.3	25.3	23.0	18.2	12.7	6.8
Mean Daily Temperature Range	5% DB	MDBR	7.5	8.1	9.2	10.8	11.2	10.1	10.1	10.0	10.3	8.8	7.9	6.4
		MCDBR	10.0	11.5	13.6	16.2	14.0	12.3	11.6	11.4	11.8	11.7	11.5	9.3
		MCWBR	8.6	9.1	8.9	9.1	7.0	5.6	4.9	4.8	5.4	6.8	7.8	7.7
	5% WB	MCDBR	9.7	11.4	12.6	14.5	12.3	10.9	10.0	9.7	10.2	10.4	10.3	9.0
MCWBR		8.7	9.5	8.9	9.2	6.8	5.7	4.9	4.9	5.4	7.0	8.1	7.9	
Clear-Sky Solar Irradiance	taub	0.301	0.317	0.341	0.371	0.390	0.438	0.440	0.429	0.370	0.352	0.310	0.291	
	taud	2.370	2.327	2.363	2.336	2.322	2.208	2.214	2.231	2.385	2.433	2.511	2.467	
	Ebn at Noon	835	880	902	896	884	838	832	829	860	831	827	822	
	Edh at Noon	80	99	108	120	125	141	139	132	105	89	70	67	
All-Sky Solar Radiation	RadAvg	1.35	2.19	3.31	4.41	5.31	5.63	5.74	5.13	4.11	2.53	1.61	1.08	
	RadStd	0.12	0.28	0.32	0.49	0.53	0.50	0.46	0.30	0.40	0.26	0.21	0.16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+85	N/A

Nomenclature: See separate page